RestoNiti

Cloud stack savings brief

Internal. Generated 2026-09-10. Compares three ways to run the same product: encrypted cloud backup + owner dashboard + QR view-only menu. Azure meters from Retail Prices API (Central India, INR) on 2026-09-10. R2 at $0.015/GB-mo, USD/INR = 88. Hours/month = 730. Customer prices excl. GST. Support/salaries not included.

How to read savings

Saving = planned Azure SQL platform minus the alternative, per month, then x12 for year. R2, OTP, and QR Pages are in all full-Premium stacks (Pages ~Rs. 0). The large gap is Azure SQL Database 1 vCore vs a small PostgreSQL. From-scratch API host is a mid estimate (Fly.io / small VM), labelled as such.

1. The three stacks (same product)

Stack	API	Database	Files / QR
A. Planned Azure (PDF Scenario B)	App Service Linux B1	SQL Gen5 1 vCore always on	R2 + Pages
B. Stay Azure, swap DB	Same Linux B1 (keep current API)	Postgres Flexible B1ms always on	R2 + Pages
C. From-scratch	Go on Fly/small VM (est.)	Same-size Postgres (B1ms price as proxy)	R2 + Pages

2. Unit prices used

Meter	Unit price	Rs./month if always on
App Service Linux B1	Rs. 1.7198/hr	Rs. 1,255
SQL Serverless Gen5	Rs. 54.8372/vCore-hr	See backup scenario (2h/day)
SQL provisioned 1 vCore	Rs. 15.9982/hr	Rs. 11,679
Postgres Flexible B1ms	Rs. 2.3409/hr	Rs. 1,709
SQL/PG storage (proxy)	Rs. 12.5261/GB-mo	5 GB backup / 15 GB dashboard
R2 storage	$0.015/GB-mo	Listed in line items

Source: https://prices.azure.com/api/retail/prices - armRegionName=centralindia, currencyCode=INR. Postgres B1ms product: Azure Database for PostgreSQL Flexible Server Burstable BS Series Compute. Storage for Postgres data disk is proxied at the SQL GP Rs./GB rate so size is comparable.

3. Backup-only platform (no owner dashboard yet)

SQL wakes ~2 hours/day at 0.5 vCore then pauses. From-scratch uses a small sleeping API + cheap Postgres.

Line item	Azure A (sleepy SQL)	From-scratch C	Calculation
API compute	Rs. 1,255	Rs. 800	Azure: 1.7198 x 730h. Scratch: mid estimate.
Database compute	Rs. 1,645	Rs. 400	Azure: 0.5 vCore x 2h x 30d x 54.8372. Scratch: sleeping PG.
DB storage 5 GB	Rs. 63	in scratch PG	5 x 12.5261
R2 + ops	Rs. 63	Rs. 63	10 GB x $0.015 x 88 + Rs. 50 ops
Misc	Rs. 400	Rs. 400	Domain/logs buffer
TOTAL / month	Rs. 3,426	Rs. 1,663	Shared platform

If you choose	You save vs Azure A / mo	You save / year
From-scratch instead of sleepy Azure SQL	Rs. 1,763	Rs. 21,158

4. Full Premium (backup + owner dashboard) - line items

Owner dashboard keeps a database busy, so Azure SQL is provisioned 1 vCore. Stack B and C use small always-on Postgres instead. OTP and R2 are the same.

Line item	A Azure SQL	B Azure+PG	C From-scratch	Calculation
API compute	Rs. 1,255	Rs. 1,255	Rs. 1,000	B1 1.7198/hr x 730. Scratch API est. Rs. 1000.
Database compute	Rs. 11,679	Rs. 1,709	Rs. 1,709	SQL 15.9982/hr vs PG B1ms 2.3409/hr x 730.
DB storage 15 GB	Rs. 188	Rs. 188	Rs. 188	15 x 12.5261
R2 backups + ops	Rs. 216	Rs. 216	Rs. 216	50 GB x $0.015 x 88 + Rs. 150
Owner auth / SMS	Rs. 800	Rs. 800	Rs. 800	Same buffer
Misc / CDN	Rs. 600	Rs. 600	Rs. 400	Scratch: Pages ~0 so lower misc
TOTAL / month	Rs. 14,738	Rs. 4,768	Rs. 4,313	Shared platform, all shops
TOTAL / year	Rs. 176,856	Rs. 57,218	Rs. 51,753	x12

5. Savings table (the headline)

Move from -> to	Save / month	Save / year	What you actually change
A Azure SQL -> B Azure+Postgres	Rs. 9,970	Rs. 119,638	Keep .NET API + R2. Change EF to Postgres. ~90% of greenfield savings.
A Azure SQL -> C from-scratch	Rs. 10,425	Rs. 125,103	Rewrite API in Go + new host. Extra save vs B is small.
B Azure+PG -> C from-scratch	Rs. 455	Rs. 5,465	Not worth a rewrite. Mostly cheaper VM vs App Service B1.

Where the money is

SQL 1 vCore always on = Rs. 11,679/mo. Postgres B1ms = Rs. 1,709/mo. Database compute difference = Rs. 9,970/mo (Rs. 119,638/year). That is the savings. Go vs .NET is not.

6. Cloud break-even at Premium Rs. 10,000/year

Fee = Rs. 833/outlet/mo. Variable cloud = Rs. 45/outlet. Break-even shops = platform / (833.33 - 45). Support time not included.

Stack	Platform / mo	Break-even Premium shops	Shops you no longer wait for vs Azure SQL
A Azure SQL	Rs. 14,738	18.7	-
B Azure+PG	Rs. 4,768	6.0	~13 fewer
C From-scratch	Rs. 4,313	5.5	~13 fewer

7. Cloud P&L at Rs. 10,000 Premium (dashboard live)

Cloud P/L/mo = (N x Rs. 833) - (platform + N x Rs. 45). Same meaning as the pricing PDF.

Stack A - planned Azure SQL

Premium N	Revenue/mo	Cloud cost/mo	Cloud P/L/mo
5	Rs. 4,167	Rs. 14,963	Rs. -10,796
10	Rs. 8,333	Rs. 15,188	Rs. -6,855
15	Rs. 12,500	Rs. 15,413	Rs. -2,913
25	Rs. 20,833	Rs. 15,863	Rs. 4,970
50	Rs. 41,667	Rs. 16,988	Rs. 24,679
100	Rs. 83,333	Rs. 19,238	Rs. 64,095

Stack B - Azure + Postgres (recommended if keeping current API)

Premium N	Revenue/mo	Cloud cost/mo	Cloud P/L/mo
5	Rs. 4,167	Rs. 4,993	Rs. -827
10	Rs. 8,333	Rs. 5,218	Rs. 3,115
15	Rs. 12,500	Rs. 5,443	Rs. 7,057
25	Rs. 20,833	Rs. 5,893	Rs. 14,940
50	Rs. 41,667	Rs. 7,018	Rs. 34,648
100	Rs. 83,333	Rs. 9,268	Rs. 74,065

Stack C - from-scratch Go + Postgres + R2

Premium N	Revenue/mo	Cloud cost/mo	Cloud P/L/mo
5	Rs. 4,167	Rs. 4,538	Rs. -371
10	Rs. 8,333	Rs. 4,763	Rs. 3,571
15	Rs. 12,500	Rs. 4,988	Rs. 7,512
25	Rs. 20,833	Rs. 5,438	Rs. 15,396
50	Rs. 41,667	Rs. 6,563	Rs. 35,104
100	Rs. 83,333	Rs. 8,813	Rs. 74,521

8. What is not a saving

Item	In all three full stacks?	Note
Cloudflare R2 backups	Yes	Do not move to S3/Blob to 'save'
QR view-only Pages	Yes	~Rs. 0; not in the SQL bill
SMS OTP buffer	Yes	Owner login; independent of Azure vs Fly
Per-outlet R2/sync share	Yes	Rs. 45/mo in P&L
Rewrite engineering months	Only stack C	Costs more than years of B vs C gap

9. Recommendation

To capture ~Rs. 9,970/month (~Rs. 119,638/year) vs the pricing PDF's Azure SQL Scenario B: keep the existing .NET API and R2, and put owner-dashboard metadata/sync on PostgreSQL (Azure Flexible B1ms or equivalent). Rewriting in Go saves only ~Rs. 455/month extra. Do not rebuild the POS for cloud cost.

Companion: docs/RESTONITI_BUSINESS_PRICING_AND_EXPENSE.pdf (customer prices). Generator: scripts/generate-cloud-stack-savings-pdf.py